

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: 3rd Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) How many different Binary Search Trees (BST) can be constructed with 4 distinct keys? [1]

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

1(b) Write the postfix expression for the following infix expression: $A + B \times C - D / E$ [1]

1(d) State the worst-case time complexity of searching an element in a Binary Search Tree (BST) of height h . [1]

$$T(n) = \mathcal{O}(h)$$

1(e) What is the maximum number of nodes in a binary tree of height h (assuming root is at height 0)? [1]

$$N_{\max} = 2^{h+1} - 1$$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Explain array storage representation in row-major order. [2.5]

$$Loc(A[i][j]) = B + (i \times N + j) \times w$$

(b) Evaluate the postfix expression: $6 \mid 2 \mid 3 \mid + \mid - \mid 3 \mid 8 \mid 2 \mid \wedge \mid + \mid * \mid 2 \mid \wedge \mid 3 \mid +$. [2.5]

--- END OF PAPER ---